# Worksheet: Constrained Optimization

ECON 441: Introduction to Mathematical Economics — Instructor: Div Bhagia

Given an objective function

$$y = f(x_1, x_2)$$

subject to the constraint

$$g(x_1, x_2) = c$$

where $c$ is a constant. We can write the Lagrange function as

$$L(x_1, x_2, \lambda) = f(x_1, x_2) + \lambda[c - g(x_1, x_2)]$$

First-order conditions:

$$\frac{\partial L}{\partial x_1} = 0, \quad \frac{\partial L}{\partial x_2} = 0, \quad \frac{\partial L}{\partial \lambda} = 0$$

For each of the problems below, set up the Lagrange function, write down the first-order conditions, and find the critical points.

*Example I*:

$$\max_{\{x_1, x_2\}} \quad x_1^2 x_2 \quad \text{s.t.} \quad 2x_1^2 + x_2^2 = 3$$

*Example II*:

$$\max_{\{x,y\}} \quad x^a y^b \quad \text{s.t.} \quad x + y = 10$$

*Example III*:

$$\max_{\{x_1,x_2\}} \quad x_1^{1/3} x_2^{2/3} \quad \text{s.t.} \quad x_1 + 4x_2 = 30$$

*Example IV*:

$$\max_{\{x,y,z\}} \quad yz + xz \quad \text{s.t.} \quad y^2 + z^2 = 1 \quad \text{and} \quad xz = 3$$